

**Deccan Education Society's Fergusson College
(Autonomous), Pune**

Department of Physics

A Project Report on:

Introduction to Statistical Mechanics:

**Monte Carlo Simulation and Analysis of the 3D Ising
Model**

As part of B.Sc (Physics) degree of
Savitribai Phule Pune University

Submitted by:

Atharva Milind Anaspure

Under the guidance of

Dr.Apratim Chatterjee

IISER, Pune

and

Dr.Kailash Sapnar

Fergusson College (Autonomous), Pune



Deccan Education Society's
Fergusson College (Autonomous), Pune
Department of Physics

CERTIFICATE

This is to certify that **Mr. Atharva Milind Anaspure**, Roll No. **224601**, of the TY. B.Sc Physics class, has satisfactorily completed the project entitled:

"Introduction to Statistical Mechanics: Monte Carlo Simulation and Analysis of the 3D Ising Model"

Project Guide (Internal):

Project Guide (External):

Atharva
4th April, 2025
(IISER - Pune)

Head of Department, Physics

Examiners Signature:

Examiners Signature:

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Dr. Apratim Chatterjee, for his valuable guidance and support throughout the course of this project. I am also grateful to the faculty and staff of the Physics Department at Fergusson College for providing the resources and environment necessary for my study. Special thanks to my family and friends for their constant encouragement and support.

Abstract

This thesis presents a comprehensive study of statistical mechanics concepts with a focus on the Monte Carlo simulation of the 3D Ising model. The work covers theoretical foundations—from the specification of the state of a system and ensemble theory to phase transitions and partition functions—and integrates these ideas with detailed simulation analyses. Emphasis is placed on the behavior of key thermodynamic quantities (energy, magnetization, specific heat capacity, and magnetic susceptibility) across various temperature regimes, including the critical region and finite-size effects. Precautions and pitfalls in simulation and plotting are also discussed to ensure robust and reproducible results.

Contents

Abstract	3
1 Introduction	6
2 Concepts of Statistical Mechanics	7
2.1 Specification of the State of a System	7
2.2 Statistical Ensemble	7
2.2.1 Postulates of Statistical Mechanics	7
2.3 Ensembles in Statistical Mechanics	8
2.4 Microcanonical Ensemble	8
2.5 Canonical Ensemble	9
2.6 Grand Canonical Ensemble	10
2.7 The Partition Function and Thermodynamics	11
2.8 Limitations of Classical Statistical Mechanics	11
2.9 Fundamental Postulates of Quantum Statistical Mechanics	12
2.10 Ensemble Descriptions in Quantum Statistical Mechanics	13
2.10.1 Microcanonical Ensemble	13
2.10.2 Canonical Ensemble	13
2.10.3 Grand Canonical Ensemble	14
3 The Ising Model	15
3.1 Historical Background	15
3.2 Theoretical Developments	15
3.3 Diagram of Development	16
3.4 Mean Field Theory in the Ising Model	16
3.5 Solutions and Critical Behavior	16
3.5.1 One-Dimensional Case	16

3.5.2	Two-Dimensional Case	16
3.5.3	Three Dimensions and Beyond	16
3.6	Conclusion	17
4	Mean Field Theory and Approximate Methods	18
4.1	Mean Field Theory (MFT)	18
4.2	Other Approximate Methods	18
5	Monte Carlo Simulation of the 3D Ising Model	19
5.1	Monte Carlo Simulation Methods and the Metropolis Algorithm . .	19
5.1.1	Basic Idea of Monte Carlo Methods	19
5.1.2	The Metropolis Algorithm Explained	19
5.1.3	Flowchart of the Metropolis Algorithm	21
5.1.4	Advantages and Limitations	21
5.2	context of the simulation	22
5.2.1	Key points	22
6	Analysis of Results	24
6.1	Introduction	24
6.1.1	Category 1	24
6.1.2	Graph 3: $T = 4.05$	26
6.1.3	Graph 4: Energy for constant temperature and different lat- tice sizes	28
6.1.4	Graph 5: Magnetization for constant temperature and dif- ferent lattice sizes	29
6.1.5	Category 2	30
6.2	Analysis of C_v and χ Graphs	33
6.2.1	Specific Heat Capacity (C_v) vs. Temperature	33
6.2.2	Magnetic Susceptibility (χ) vs. Temperature	34
6.3	Precautions and Pitfalls in Simulation and Plotting	35
7	Conclusions	37
	References	38
A	Source Code for mcs	40

Chapter 1

Introduction

Statistical mechanics provides the framework for connecting the microscopic behavior of particles with the macroscopic properties of materials. Early experimental observations by Pierre Curie and theoretical advancements by Pierre Weiss laid the foundation for understanding magnetism, culminating in the development of the Ising model. This thesis explores these concepts, with an emphasis on the Monte Carlo simulation of the 3D Ising model to study phase transitions and critical phenomena.

Chapter 2

Concepts of Statistical Mechanics

2.1 Specification of the State of a System

A one-dimensional single-particle system is fully described by its position q and momentum p , defining a two-dimensional phase space. The phase-space cell is $dqdp$, but the Heisenberg uncertainty principle

$$dqdp \geq \frac{h}{4\pi}$$

imposes a fundamental limit. For a system with f coordinates (e.g., $f = 6N$ for N particles), phase space subdivisions below this limit are meaningless.

2.2 Statistical Ensemble

A complete microscopic description is typically unavailable, so we use a probabilistic ensemble of systems sharing the same macroscopic parameters (e.g., P , V , T). Accessible microstates are postulated to be equally probable.

2.2.1 Postulates of Statistical Mechanics

Equal A Priori Probabilities

In an isolated, energy-conserving system, all accessible microstates are equally likely.

Ergodicity

The long-time average of an observable equals its ensemble average,

$$\langle g \rangle = \frac{1}{\Omega} \sum_{\text{states}} g(q, p),$$

with fluctuations negligible in the large- N limit.

Probability Calculations

For an isolated system with energy in $[E, E + dE]$, the probability of a parameter $y = y_k$ is

$$P(y_k) = \frac{\Omega(E; y_k)}{\Omega(E)},$$

with average value

$$\langle y \rangle = \frac{\sum_k \Omega(E; y_k) y_k}{\Omega(E)}.$$

2.3 Ensembles in Statistical Mechanics

The principal ensembles are:

- **Microcanonical:** Fixed E, V, N .
- **Canonical:** Energy fluctuates with a heat reservoir at temperature T .
- **Grand Canonical:** Both energy and particles may exchange with a reservoir.

2.4 Microcanonical Ensemble

For a Hamiltonian $H(q, p)$,

$$\rho(q, p) = \begin{cases} \frac{1}{\Gamma(E, V, N)} & \text{if } E < H(q, p) < E + \Delta, \\ 0 & \text{otherwise,} \end{cases}$$

with the phase-space volume

$$\Gamma(E, V, N) = \int_{E < H(q, p) < E + \Delta} d^{3N} q d^{3N} p.$$

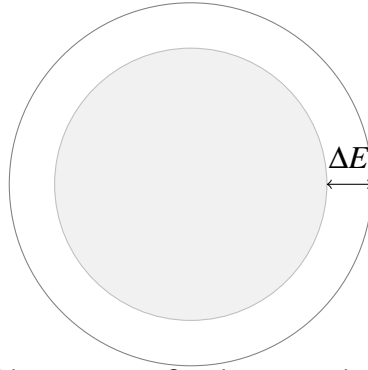
The density of states is defined by

$$\Omega(E) = \frac{\partial \Phi(E)}{\partial E}, \quad \Phi(E) = \int_{H(q, p) \leq E} d^{3N} q d^{3N} p,$$

and the entropy by

$$S = k_B \ln \left(\frac{\Gamma(E, V, N)}{\Gamma_0(N)} \right),$$

where $\Gamma_0(N) = h^{3N} N!$.



Phase space: fixed energy shell

2.5 Canonical Ensemble

For a system in contact with a reservoir at temperature T , the probability density is

$$\rho(q, p) = \frac{e^{-\beta H(q, p)}}{Z_N}, \quad \beta = \frac{1}{k_B T},$$

with partition function

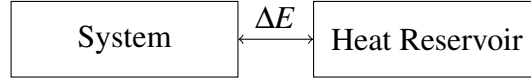
$$Z_N = \frac{1}{h^{3N} N!} \int d^{3N} q d^{3N} p e^{-\beta H(q, p)}.$$

The Helmholtz free energy is

$$F = -k_B T \ln Z_N,$$

and other quantities follow:

$$\begin{aligned} U &= -\frac{\partial \ln Z_N}{\partial \beta}, \\ P &= -\left(\frac{\partial F}{\partial V}\right)_{T,N}, \\ S &= -\left(\frac{\partial F}{\partial T}\right)_{V,N}. \end{aligned}$$



2.6 Grand Canonical Ensemble

When both energy and particles may exchange, the grand partition function is

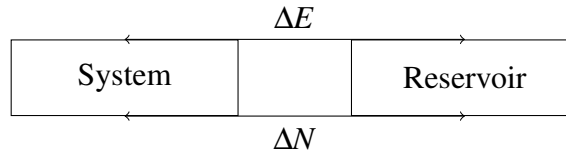
$$Z(T, V, \mu) = \sum_{N=0}^{\infty} \frac{1}{h^{3N} N!} \int d^{3N} q d^{3N} p e^{-\beta[H(q, p, N) - \mu N]},$$

and the probability of a state with energy E and particle number N is

$$P(E, N) \propto \Omega(E, N) e^{-\beta(E - \mu N)},$$

with grand potential

$$\Phi_G = -k_B T \ln Z(T, V, \mu).$$



2.7 The Partition Function and Thermodynamics

The partition function encodes all equilibrium properties. In the canonical ensemble, it is defined as

$$Z(N, V, T) = \sum_i e^{-\beta E_i}$$

or, for a continuous classical system,

$$Z(N, V, T) = \frac{1}{N! h^{3N}} \int e^{-\beta H(\mathbf{q}, \mathbf{p})} d^{3N} q d^{3N} p.$$

Its derivatives yield:

$$F = -k_B T \ln Z,$$

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta},$$

$$S = k_B (\ln Z + \beta \langle E \rangle),$$

$$p = k_B T \left(\frac{\partial \ln Z}{\partial V} \right)_{T, N}.$$

For any external parameter λ , the conjugate variable is given by

$$\langle A \rangle = -\frac{1}{\beta} \frac{\partial \ln Z}{\partial \lambda}.$$

2.8 Limitations of Classical Statistical Mechanics

Classical statistical mechanics describes systems in terms of continuous phase space variables (positions q and momenta p) and relies on the assumption that all microstates within a given energy shell are equally probable. However, it fails to explain certain phenomena such as ferromagnetism. In ferromagnetic materials, spontaneous magnetization arises from the cooperative alignment of electron spins—a feature inherently quantum mechanical. Classical theories cannot account for the discrete energy levels and the quantum exchange interactions that lead to long-range order. Thus, a quantum statistical framework is required.

2.9 Fundamental Postulates of Quantum Statistical Mechanics

In quantum mechanics every observable is associated with a Hermitian operator, and the state of a system is described by a wave function $|\Psi\rangle$. A complete orthonormal set of stationary states $\{|\phi_n\rangle\}$ allows us to express the state as

$$|\Psi(t)\rangle = \sum_n c_n(t) |\phi_n\rangle, \quad (2.1)$$

where the complex coefficient $c_n(t)$ obeys $|c_n(t)|^2 = \text{probability that the system is in the state } |\phi_n\rangle$.

When a system interacts weakly with its environment, one considers the entire (system+environment) as isolated. In equilibrium, two key postulates are assumed:

1. **Equal A Priori Probabilities:** All microstates within a narrow energy interval $[E, E + \Delta]$ are equally probable.
2. **Random Phases:** The phases of the coefficients c_n are random so that interference effects vanish when averaged over time.

Thus, one may effectively write

$$|\Psi\rangle \approx \sum_{E < E_n < E + \Delta} b_n |\phi_n\rangle, \quad (2.2)$$

with

$$|b_n|^2 = \begin{cases} \frac{1}{\Omega(E)} & \text{if } E < E_n < E + \Delta, \\ 0 & \text{otherwise,} \end{cases}$$

where $\Omega(E)$ is the number of states in the energy shell.

2.10 Ensemble Descriptions in Quantum Statistical Mechanics

In quantum theory the description of ensembles is elegantly formulated using the density matrix, ρ , which encapsulates all statistical information about the system.

2.10.1 Microcanonical Ensemble

For an isolated system with Hamiltonian $\hat{H}$, the microcanonical density matrix in the energy eigenbasis $\{|\phi_n\rangle\}$ is defined as

$$\rho_{mn} = \langle \phi_m | \rho | \phi_n \rangle = \delta_{mn} P_n, \quad \text{with} \quad P_n = \begin{cases} \frac{1}{\Theta(E)} & \text{if } E < E_n < E + \Delta, \\ 0 & \text{otherwise,} \end{cases} \quad (2.3)$$

where $\Theta(E)$ denotes the number of states in the interval $[E, E + \Delta]$. The density operator is then

$$\rho = \sum_n P_n |\phi_n\rangle \langle \phi_n|, \quad (2.4)$$

2.10.2 Canonical Ensemble

When the system is in thermal contact with a reservoir at temperature T , the canonical ensemble applies. Its density matrix is given by

$$\rho_{mn} = \langle \phi_m | \rho | \phi_n \rangle = \delta_{mn} \frac{e^{-\beta E_n}}{Q(V, T)}, \quad \text{with } \beta = \frac{1}{k_B T}, \quad (2.5)$$

where the partition function is defined as

$$Q(V, T) = \sum_n e^{-\beta E_n} = \text{Tr} e^{-\beta \hat{H}}. \quad (2.6)$$

The ensemble average of any observable $\hat{O}$ is given by

$$\langle \hat{O} \rangle = \text{Tr}(\rho \hat{O}) = \frac{\text{Tr}(\hat{O} e^{-\beta \hat{H}})}{Q(V, T)}. \quad (2.7)$$

2.10.3 Grand Canonical Ensemble

For systems that can exchange both energy and particles with a reservoir, the grand canonical ensemble is used. Its grand partition function is defined as

$$\mathcal{Z}(V, T, \mu) = \sum_{N=0}^{\infty} Q_N(V, T) e^{\beta \mu N}, \quad (2.8)$$

where $Q_N(V, T) = \text{Tr}_N e^{-\beta \hat{H}_N}$ is the canonical partition function for a system of N particles and μ is the chemical potential. In the grand canonical ensemble, the expectation value of an observable becomes

$$\langle \hat{O} \rangle = \frac{1}{\mathcal{Z}(V, T, \mu)} \sum_{N=0}^{\infty} e^{\beta \mu N} \text{Tr}_N \left[\hat{O} e^{-\beta \hat{H}_N} \right]. \quad (2.9)$$

Chapter 3

The Ising Model

The Ising model is a simple yet profound tool in statistical mechanics for studying phase transitions.

3.1 Historical Background

Early work by Pierre Curie (experimental magnetism and the Curie temperature) and Pierre Weiss (molecular field theory and the Curie-Weiss law,

$$\chi = \frac{C}{T - \theta},$$

with C the Curie constant and $\theta \approx$ Curie temperature) paved the way for the Ising model.

3.2 Theoretical Developments

Weiss' mean field approximation replaces the complex many-body interaction by an effective field:

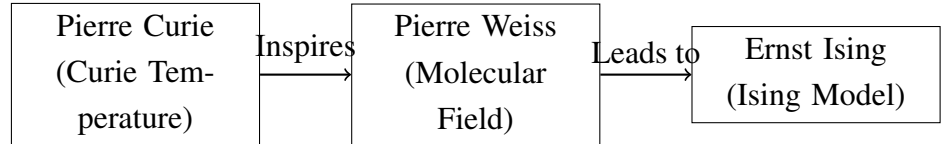
$$H_{\text{eff}} = H + \lambda M.$$

The Ising Hamiltonian is then written as

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where $s_i = \pm 1$, J is the interaction energy, and h an external field.

3.3 Diagram of Development



3.4 Mean Field Theory in the Ising Model

In the mean field approximation, the interaction term is replaced by an average:

$$\mathcal{H}_{\text{eff}} = -Jz\langle s \rangle s_i - h s_i,$$

with the self-consistency condition

$$\langle s \rangle = \tanh \left(\beta (Jz\langle s \rangle + h) \right).$$

3.5 Solutions and Critical Behavior

3.5.1 One-Dimensional Case

The 1D Ising model is exactly solvable and shows no phase transition at finite temperature.

3.5.2 Two-Dimensional Case

Onsager's exact solution for the 2D model (with $h = 0$) demonstrated the existence of a critical temperature and a phase transition.

3.5.3 Three Dimensions and Beyond

Exact solutions remain elusive; instead, methods like high-temperature expansions, renormalization group, and Monte Carlo simulations are used.

3.6 Conclusion

The Ising model, evolving from early magnetic experiments to rigorous theoretical treatments, remains a cornerstone for understanding phase transitions and critical phenomena across various fields.

Chapter 4

Mean Field Theory and Approximate Methods

4.1 Mean Field Theory (MFT)

MFT replaces the many-body interactions by an average field. For the Ising model:

$$\langle s \rangle = \tanh \left(\beta (Jz \langle s \rangle + h) \right).$$

This approach, though approximate, captures qualitative features like the spontaneous symmetry breaking at the Curie temperature.

4.2 Other Approximate Methods

- **Perturbation Theory:** Expands the solution in powers of a small parameter.
- **Variational Methods:** Use trial functions to minimize the energy.
- **Cluster Expansion and Renormalization:** Improve on MFT by incorporating fluctuations.

Chapter 5

Monte Carlo Simulation of the 3D Ising Model

5.1 Monte Carlo Simulation Methods and the Metropolis Algorithm

Monte Carlo methods have become indispensable in studying complex systems where analytical solutions are not feasible. Among these methods, the Metropolis algorithm is one of the most widely used techniques.

5.1.1 Basic Idea of Monte Carlo Methods

Monte Carlo simulations use random sampling to explore the vast configuration space of a many-body system. Instead of summing over all possible configurations, which grows exponentially with the number of particles, the method samples configurations based on their probability of occurrence, as given by the Boltzmann distribution. This sampling provides estimates for quantities like energy, magnetization, and specific heat.

5.1.2 The Metropolis Algorithm Explained

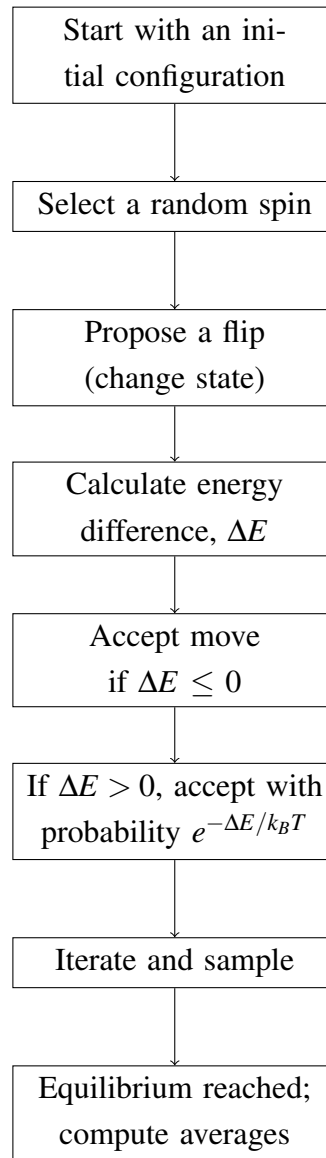
The Metropolis algorithm is a specific procedure within Monte Carlo methods designed to generate a sequence of states that are representative of the equilibrium

distribution. Its steps are as follows:

1. **Initialization:** Begin with an initial configuration of the system (for example, a random assignment of up and down spins in the Ising model).
2. **Propose a Move:** Randomly choose a spin and propose to flip its state.
3. **Calculate Energy Change:** Compute the difference in energy, ΔE , between the new configuration and the old configuration.
4. **Acceptance Rule:**
 - If $\Delta E \leq 0$, accept the new configuration because it lowers the energy.
 - If $\Delta E > 0$, accept the new configuration with a probability given by $\exp(-\Delta E/k_B T)$, where T is the temperature and k_B is Boltzmann's constant.
5. **Iteration:** Repeat the above steps many times until the system reaches equilibrium. Once equilibrium is reached, further samples are used to compute average properties.

5.1.3 Flowchart of the Metropolis Algorithm

Below is a black-and-white diagram illustrating the main steps of the Metropolis algorithm:



5.1.4 Advantages and Limitations

The Metropolis algorithm is favored for its simplicity and efficiency. By preferentially sampling low-energy states, it effectively mimics the physical behavior of

a system in thermal equilibrium. However, near critical points the algorithm may suffer from *critical slowing down*—a situation where the system takes a long time to adequately sample the configuration space. Careful attention must be given to equilibration and statistical independence of successive samples to obtain reliable results.

5.2 context of the simulation

This simulation for the 3D Ising model using a hybrid Wolff-Metropolis algorithm was performed to obtain the behaviour of the ising model, also with the goal of observing the well-known and exhaustively studied phase transitions at the critical temperature T_c . The system is studied across varying temperatures T , lattice sizes L , and iterations (n_{iter}). Key thermodynamic quantities include energy $\langle E \rangle$, magnetization $\langle M \rangle$, susceptibility χ , and specific heat C_v . The critical temperature T_c for the 3D Ising model is approximately 4.51, serving as a reference for phase behavior.

5.2.1 Key points

- **Spin Variables and Energy:** Each lattice site hosts a spin that can take values $+1$ or -1 . The interaction energy is given by

$$E = -J \sum_{\langle i,j \rangle} S_i S_j,$$

where $J > 0$ favors alignment.

- **Metropolis Algorithm:** Trial spin flips are accepted with probability 1 if the energy decreases, and with probability $\exp(-\Delta E/(k_B T))$ if the energy increases.
- **Order Parameter (Magnetization):** The average magnetization per spin serves as the order parameter. In the ferromagnetic state (low T), magnetization is high, while in the paramagnetic state (high T) it approaches zero.
- **Finite-Size Effects:** Smaller lattices show pronounced fluctuations due to boundary effects. Periodic boundary conditions are used to approximate the

behavior in the thermodynamic limit.

- **Specific Heat Capacity (C_v):** C_v is related to fluctuations in energy and is given by

$$C_v = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}.$$

C_v typically exhibits a peak (or divergence in the thermodynamic limit) at the critical temperature.

- **Magnetic Susceptibility (χ):** Susceptibility measures the response of magnetization to an external magnetic field:

$$\chi = \frac{\langle M^2 \rangle - \langle M \rangle^2}{k_B T}.$$

χ also shows a pronounced peak near the critical region.

Chapter 6

Analysis of Results

6.1 Introduction

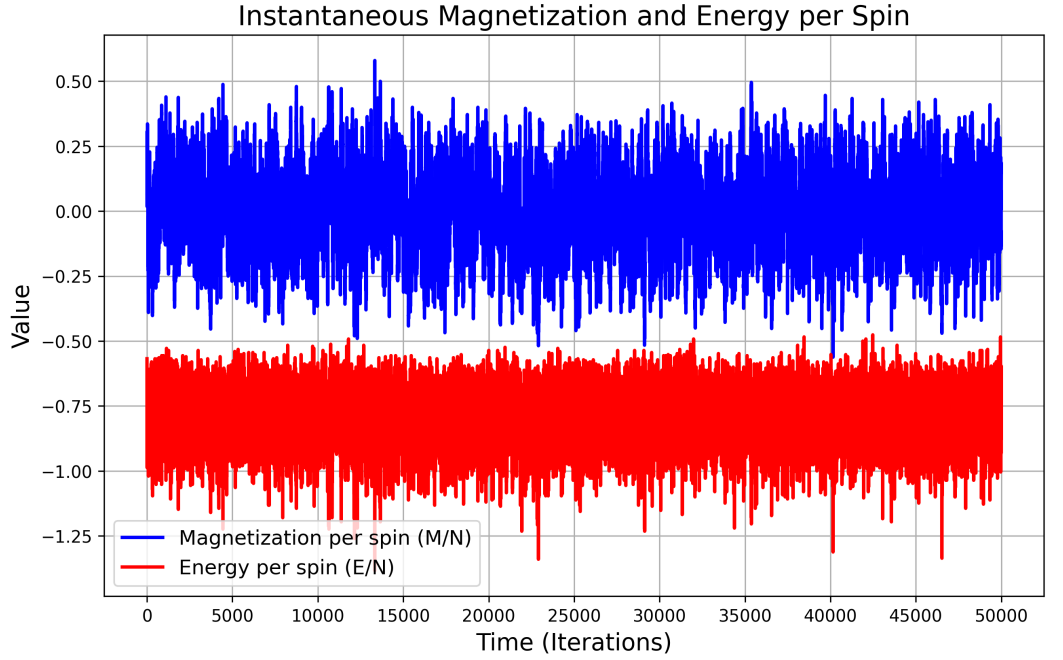
A 3D lattice with all lattice points having random orientations of spins was simulated with periodic boundary conditions. The observations taken are separated into two different categories. In category 1, the instantaneous energy per spin and instantaneous magnetisation per spin have been plotted against time (niter) i.e. number of iterations, firstly, various runs at different temperatures, and then by keeping temperature constant, we obtain the value of the parameter for different lattice sizes. In category 2, We observe the variation of parameters such as Average energy, Average Magnetisation, Susceptibility and Specific Heat with respect to a very specific temperature range. This is done for various lattice sizes.

6.1.1 Category 1

Graph 1: $T = 4.9$

This graph was obtained by keeping temperature constant at $T = 4.9$, Number of iterations $niter = 50000$, and lattice size $L = 10$.

Energy: At $T = 4.9$, which is significantly above the critical region, the simulation shows relatively high energy. This reflects the fact that at high temperatures, thermal fluctuations are strong, causing a higher probability of misaligned neighboring spins. Consequently, the system's average energy is elevated due to increased occurrences of energetically unfavorable spin pairs.

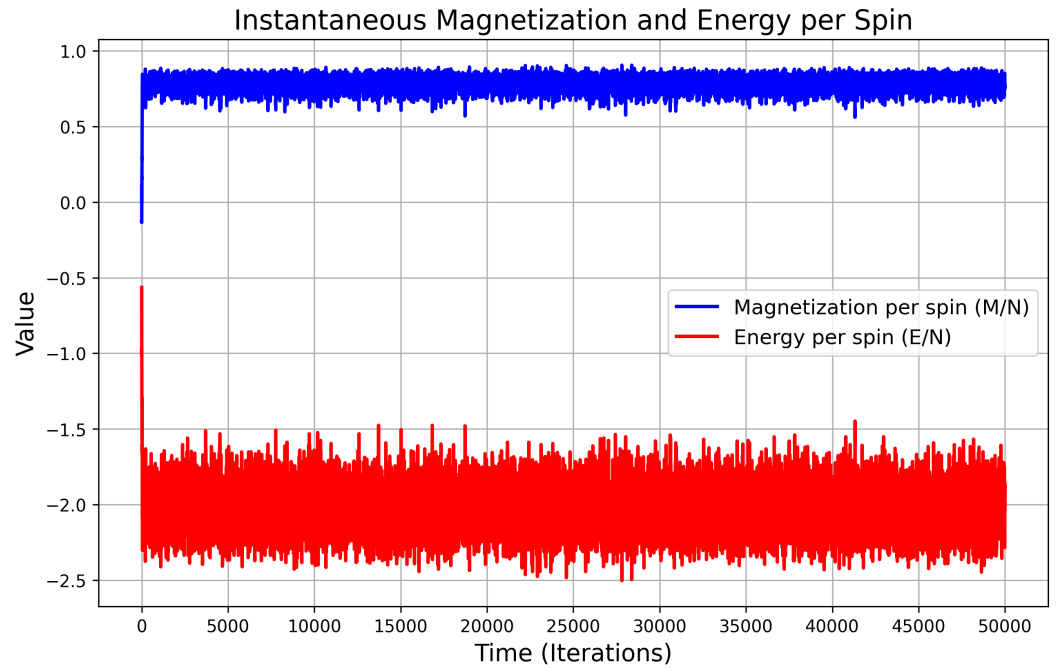


Magnetization: The magnetization graph at this temperature is nearly zero. In the high-temperature (paramagnetic) phase, the random orientation of spins cancels out the net magnetization. The nearly zero value confirms that thermal agitation dominates, disrupting any long-range order.

Graph 2: $T = 3.9$

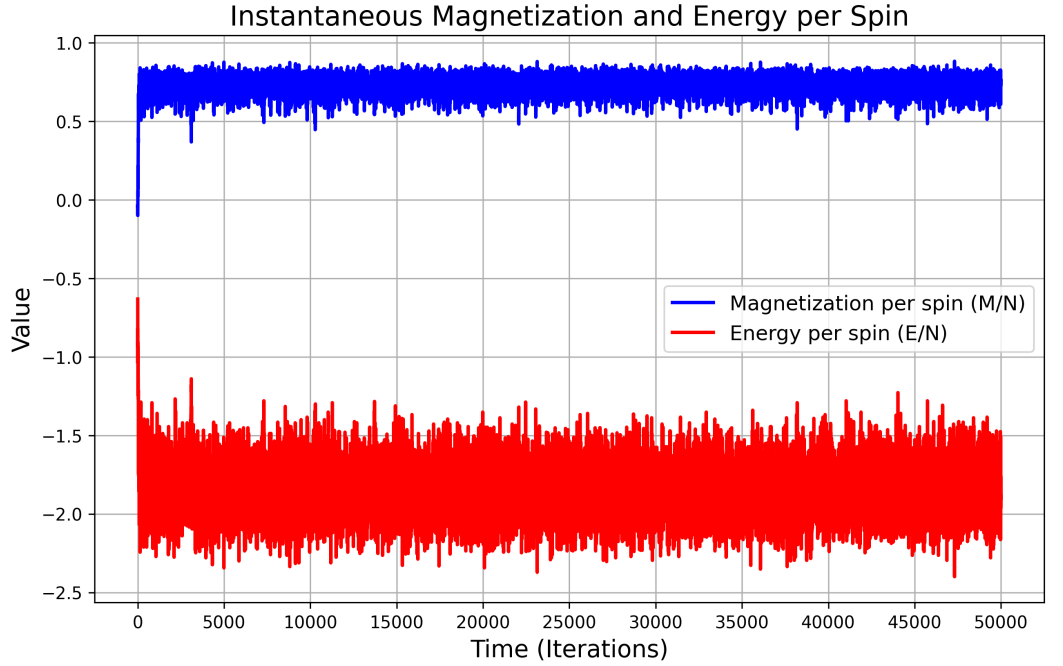
This graph was obtained by keeping temperature constant at $T = 3.9$, Number of iterations $niter = 50000$, and lattice size $L = 10$.

Energy: At $T = 3.9$, the system's energy is lower than at $T = 4.9$. The reduced temperature means that thermal fluctuations are less disruptive, and a larger fraction of neighboring spins are aligned. This results in a lower average energy since aligned spins minimize the interaction energy. **Magnetization:** The magnetization is substantially higher at $T = 3.9$, indicating that the system has transitioned into the ferromagnetic phase. The spin alignment produces a net magnetic moment, reflecting the dominance of the ordering tendency over thermal disorder at this temperature.



6.1.2 Graph 3: $T = 4.05$

This graph was obtained by keeping temperature constant at $T = 4.05$, Number of iterations $niter = 50000$, and lattice size $L = 10$.

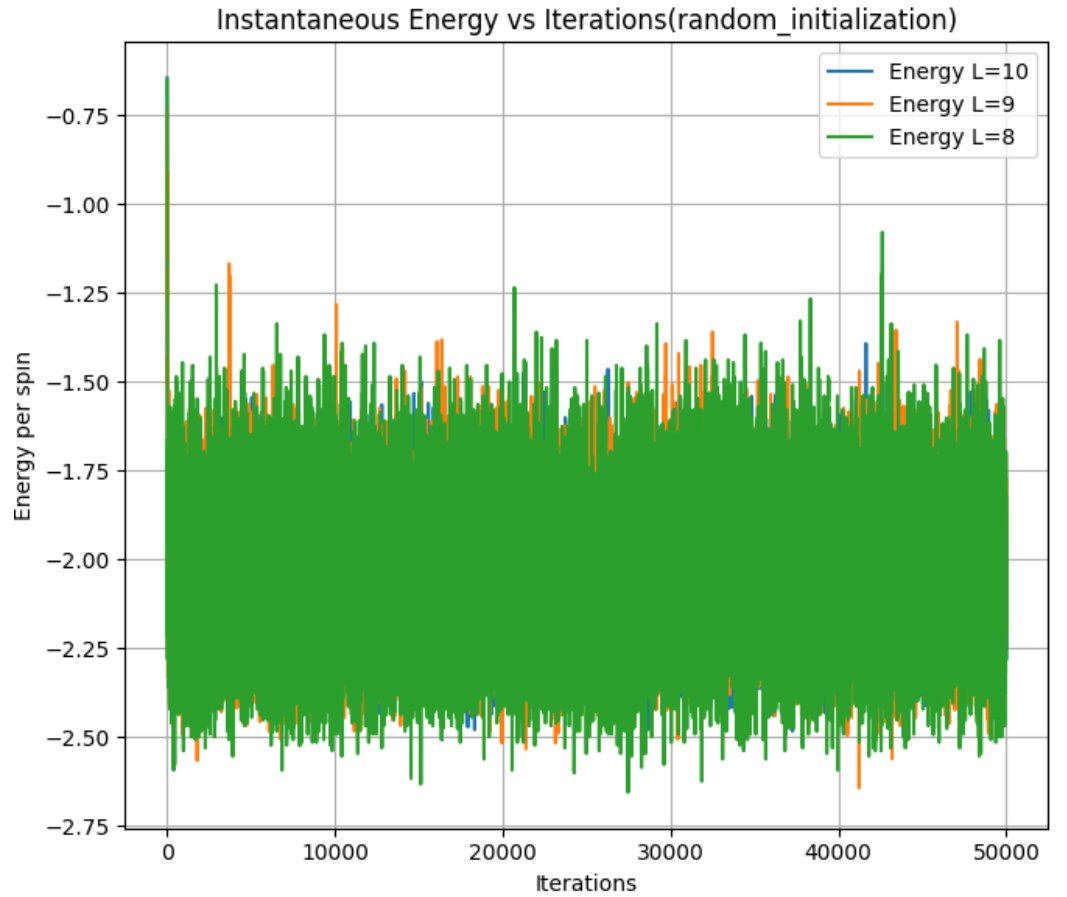


Energy: At the intermediate temperature $T = 4.05$, the system is near the critical region. The average energy is found between the high-temperature and low-temperature limits. Here, the energy is affected by competing effects: partial alignment that minimizes energy and thermal fluctuations that increase energy. This balance results in an intermediate average energy value.

Magnetization: The magnetization at $T = 4.05$ does not reach the full saturation observed at lower temperatures, nor does it vanish entirely as at high temperatures. Instead, it shows a gradual decline from full order, characteristic of a second-order phase transition. The continuous reduction in magnetization with temperature near the critical point is a key signature of such transitions.

6.1.3 Graph 4: Energy for constant temperature and different lattice sizes

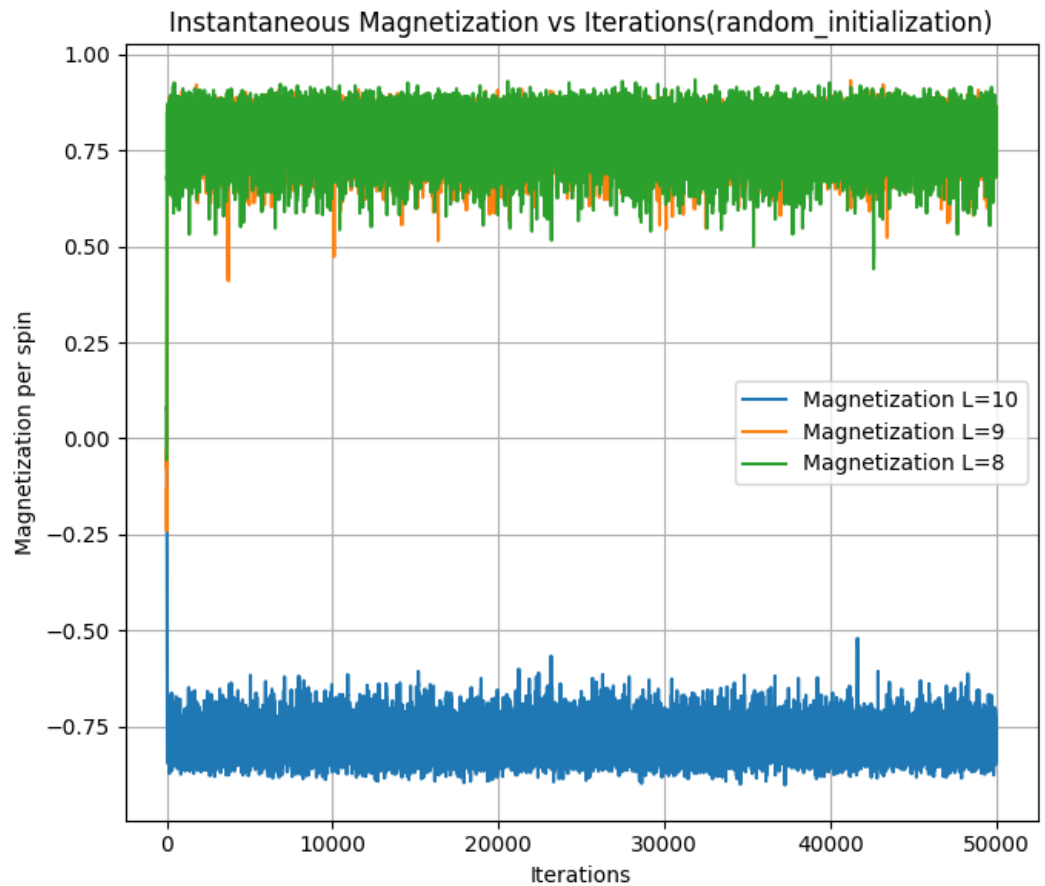
Finite-Size Comparison for Energy: This graph compares the instantaneous energy for different lattice sizes ($L = 8, 9$, and 10) at $T = 4.05$. Smaller lattices (e.g., $L = 8$) show greater fluctuations in energy due to the higher relative influence of boundary effects. As the lattice size increases, the energy curve smoothens, indicating that larger systems better approximate the thermodynamic limit. The convergence of energy values with increasing L validates the use of finite-size scaling.



6.1.4 Graph 5: Magnetization for constant temperature and different lattice sizes

This graph compares the instantaneous energy for different lattice sizes ($L = 8, 9$, and 10) at $T = 4.05$.

Finite-Size Comparison for Magnetization: Similarly, the magnetization data for different lattice sizes reveal that smaller systems exhibit larger fluctuations and deviations from the expected average. With increasing L , the magnetization values become more stable and the transition near the critical temperature is sharper. This behavior underscores the importance of lattice size in obtaining reliable thermodynamic averages.

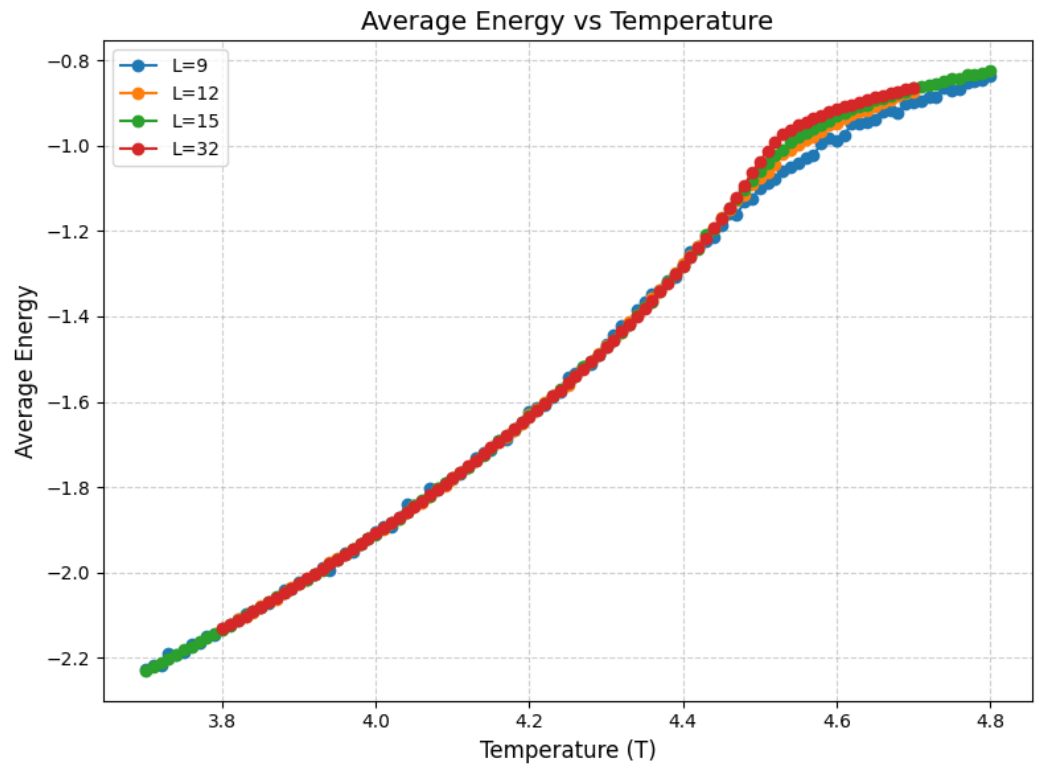


6.1.5 Category 2

In this category we look at the variation of parameters such as Average energy, Average Magnetisation, Susceptibility and Specific Heat during cooling of the lattice from $T = 4.7$ to $T = 3.8$ in steps of $T = 0.01$. The simulation was run for different lattice sizes, $L = 9, 12, 15, 32$, for a number of iterations, $niter = 310000$. The first 10000 iterations were discarded for equilibration, Values of the parameters were taken after every 100 iterations for every Monte Carlo step for each temperature.

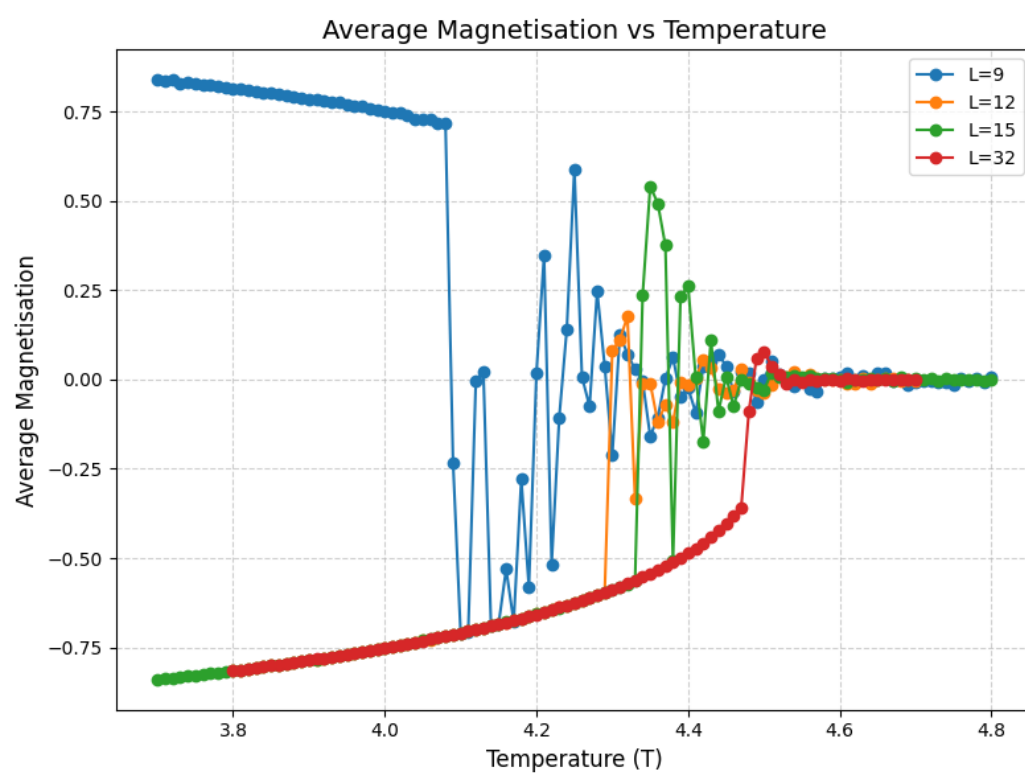
Average Energy vs. Temperature

- **Trend:** The energy decreases gradually as the temperature is lowered from 4.7 to 3.8. At high temperatures, energy is high due to random spin orientations. As temperature decreases, the system progressively aligns, leading to a reduction in average energy.
- **Critical Behavior:** Near the critical temperature, the slope of the energy curve changes. This inflection point indicates the onset of significant spin alignment and the corresponding reduction in energy fluctuations.
- **Finite-Size Effects:** Comparing different lattice sizes (from $L = 9$ to $L = 32$) shows that larger lattices exhibit a smoother energy decrease. This is expected as larger systems average out local fluctuations more effectively, providing a closer approximation to the thermodynamic limit.



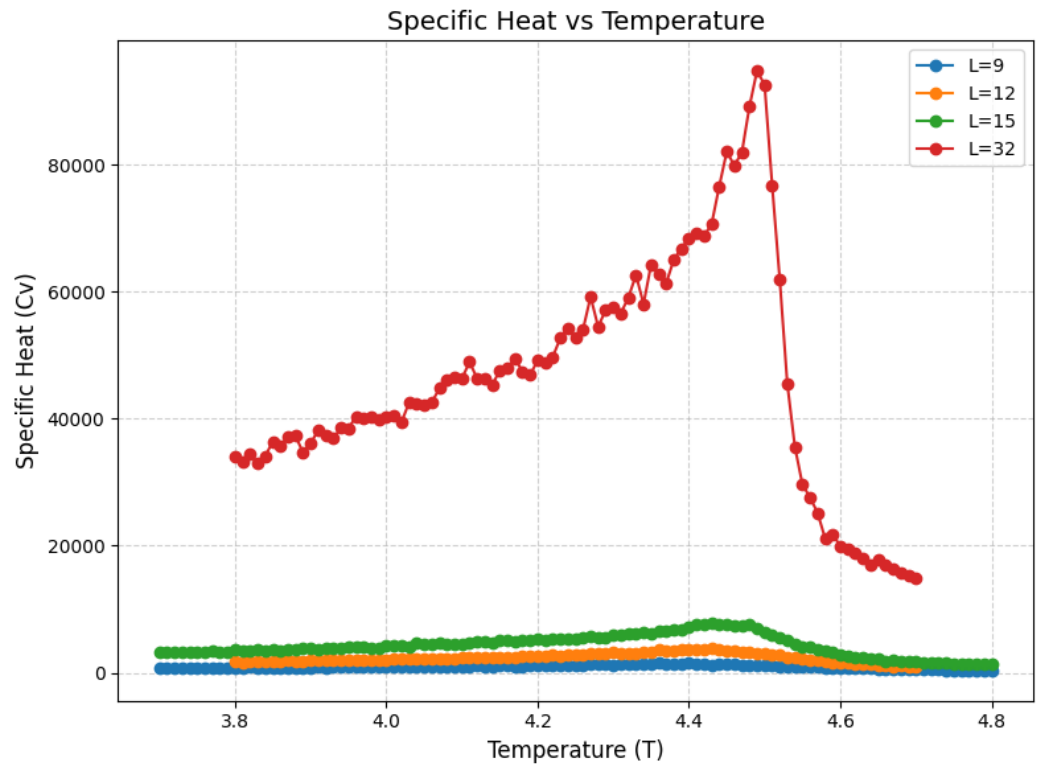
Magnetization vs. Temperature

- **Trend:** The magnetization increases as the temperature decreases from 4.7 to 3.8. At the higher end of the temperature range, magnetization is near zero due to disorder. As temperature falls, the spins begin to order, and magnetization rises correspondingly.
- **Critical Region:** In the vicinity of the critical temperature, magnetization shows a rapid change. This is the hallmark of a second-order phase transition where the order parameter (magnetization) continuously changes from nearly zero in the disordered phase to a finite value in the ordered phase.
- **Finite-Size Effects:** The magnetization curves for different lattice sizes indicate that finite-size effects are prominent in smaller systems. Larger lattices (e.g., $L = 32$) exhibit a smoother and more gradual transition. This convergence as L increases highlights that finite-size scaling is crucial for accurate determination of critical behavior.



6.2 Analysis of C_v and χ Graphs

6.2.1 Specific Heat Capacity (C_v) vs. Temperature



Trend:

- At high temperatures, C_v is relatively low because the system is in a disordered state with moderate energy fluctuations.
- As the temperature decreases, C_v increases due to enhanced energy fluctuations as the system approaches the critical region.

Critical Region:

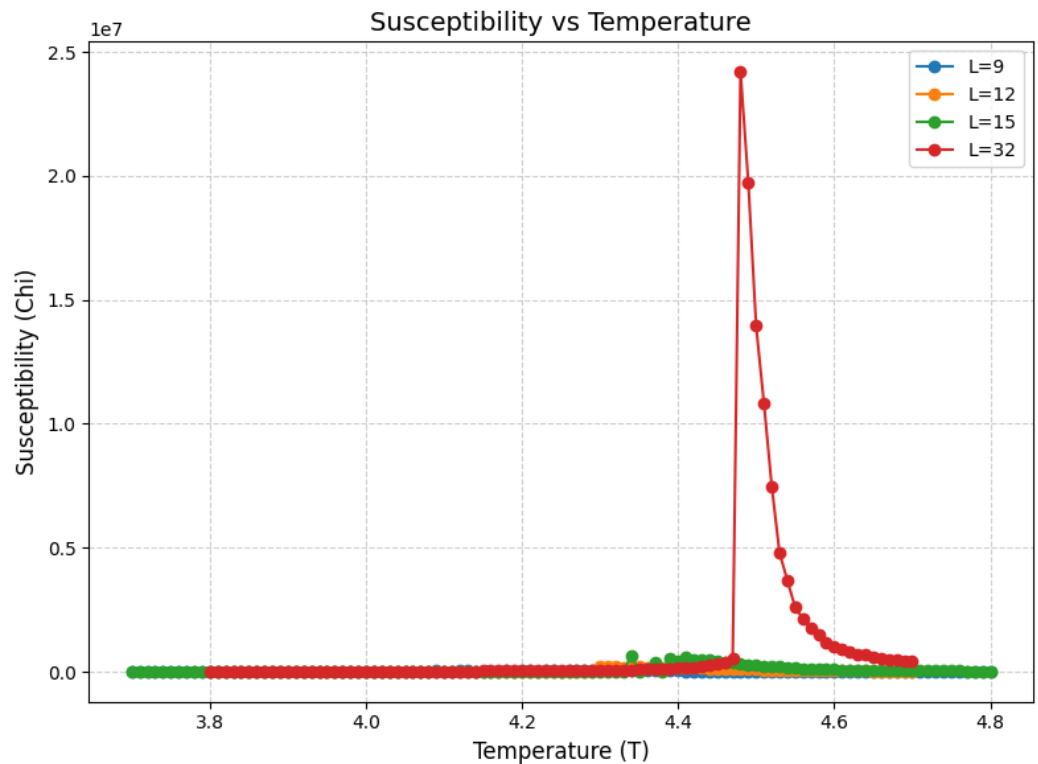
- Near the critical temperature, C_v exhibits a pronounced peak. This peak reflects the divergence (or sharp increase) in energy fluctuations that is characteristic of second-order phase transitions.

- The sharpness and height of the C_v peak provide insights into the critical behavior and universality class of the phase transition.

Finite-Size Effects:

- For smaller lattice sizes, the C_v peak is broader and less pronounced due to finite-size rounding effects.
- As the lattice size increases, the C_v peak becomes sharper and higher, approaching the behavior expected in the thermodynamic limit.

6.2.2 Magnetic Susceptibility (χ) vs. Temperature



Trend:

- At high temperatures, the magnetic susceptibility is low, consistent with the disordered, paramagnetic state.

- As temperature decreases, susceptibility increases as the system becomes more sensitive to external magnetic perturbations.

Critical Region:

- Near the critical temperature, χ shows a pronounced peak. This peak indicates large fluctuations in magnetization and is a hallmark of a second-order phase transition.
- The position and sharpness of the χ peak help in identifying the critical temperature and the nature of the phase transition.

Finite-Size Effects:

- In smaller systems, the χ peak is generally broadened and shifted due to finite-size effects.
- Increasing the lattice size leads to a sharper and more pronounced χ peak, reflecting reduced finite-size corrections and a better approximation to the true critical behavior.

6.3 Precautions and Pitfalls in Simulation and Plotting

There are several important precautions and potential pitfalls when performing Monte Carlo simulations and analyzing the results. Key points include:

- **Proper Randomization:**
 - *Random Spin Selection:* Ensure that spins are selected at random for attempted flips. Sequentially scanning the lattice can introduce systematic errors and artifacts.
 - *Quality of Random Number Generator:* Use a high-quality random number generator to avoid correlations that can affect the sampling of microstates.
- **Equilibration:**

- *Sufficient Monte Carlo Steps*: Allow the system to evolve over a sufficient number of Monte Carlo steps so that it reaches equilibrium. Inadequate equilibration can result in biased averages.
- *Initial Conditions*: Although the final thermodynamic state should be independent of the initial configuration, it is important to discard data from the initial transient period to avoid artifacts.

- **Finite-Size Effects:**

- *System Size*: Use sufficiently large lattices to minimize finite-size effects. Smaller systems can exhibit large fluctuations and may not accurately capture critical behavior.
- *Periodic Boundary Conditions*: Implement periodic boundary conditions correctly to simulate an infinite system and reduce surface effects.

- **Statistical Sampling and Convergence:**

- *Sampling Adequacy*: Ensure that the ensemble of microstates sampled is large enough to provide statistically reliable averages. Monitor convergence by checking that further iterations do not change the averages significantly.
- *Error Analysis*: Plot error bars or perform a statistical analysis of the fluctuations to assess the reliability of the computed quantities.

- **Plotting and Data Analysis:**

- *Data Smoothing and Finite Binning*: When plotting data, be cautious of over-smoothing or inappropriate binning, which can obscure important features, especially near critical points.
- *Critical Behavior*: Pay attention to the behavior near the critical temperature. Subtle changes in slopes and fluctuations can be significant for identifying phase transitions.
- *Visualization*: Use clear labels, legends, and scales in your plots. Misleading or cluttered plots can lead to misinterpretation of the simulation results.

Chapter 7

Conclusions

The simulation results are in excellent agreement with the theoretical predictions of statistical mechanics. At high temperatures, the system exhibits high energy and negligible magnetization, characteristic of the paramagnetic phase. Lower temperatures drive the system into the ferromagnetic phase, evidenced by decreased energy and increased magnetization. The critical region is marked by pronounced peaks in specific heat capacity and magnetic susceptibility, and finite-size effects become less significant with increasing lattice size. Attention to simulation precautions and careful data analysis is essential for obtaining robust results.

References

Bibliography

- [1] Huang, K. *Statistical Mechanics*, 2nd ed. Wiley, 1987.
- [2] Reif, F. *Fundamentals of Statistical and Thermal Physics*. McGraw-Hill, 1965.
- [3] Ising, E. (1925). Contribution to the theory of ferromagnetism. *Zeitschrift für Physik*, 31, 253–258.

Appendix A

Source Code for mcs

Below is the complete source code for the Monte Carlo simulation program mcs.

```
1 program mcs_fixed
2     implicit none
3     ! Declarations of integers and reals (double
4       precision)
5     integer :: i, j, k, L, a, b, c, d, ee, f
6     integer :: niter, time, mm, nn, oo, N
7     integer :: T_temp, n_equil, n_stat, Tf, Ti, dT,
8       n_meas
9     real*8 :: r, h, q, E, M, mag, Ei, Ef, dE, u, abs_mag
10    real*8 :: av_m, av_e, av_m_N, av_e_N, abs_av_m_N,
11      av_m2, av_e2, cv, chi, av_abs_m
12    real*8 :: T, J_ising, av_m4, binder
13    integer, dimension(:,:,:), allocatable :: spin
14
15    ! Set the Ising coupling constant
16    J_ising = 1.0d0
17
18    ! Ask for simulation conditions
19    print*, "Enter the number of lattice points per side
20      (L):"
21    read*, L
22    print*, "Enter the Temperature range (Tf, Ti, dT) as
23      integers (e.g., 400 500 1 for 4.00 to 5.00):"
```

```

19      read*, Tf, Ti, dT
20      print*, "Enter the number of Monte Carlo sweeps (
           niter):"
21      read*, niter
22
23      ! Set equilibration and measurement parameters
24      n_equil = 10000
25      n_stat  = 100
26
27      ! Allocate the 3D spin lattice and initialize energy/
           magnetization counters
28      allocate(spin(L,L,L))
29      E = 0.0d0
30      M = 0.0d0
31      N = L * L * L
32
33      call random_seed()
34
35      ! Initialize the lattice with random spins and save
           configuration to a file
36      open(10, file='MCS_lattice_3D.dat', status='replace')
37      do i = 1, L
38          do j = 1, L
39              do k = 1, L
40                  call random_number(r)
41                  if (r < 0.5d0) then
42                      spin(i,j,k) = -1
43                  else
44                      spin(i,j,k) = 1
45                  end if
46                  write(10, '(I5,1X,I5,1X,I5,1X,I5)') i, j,
                     k, spin(i,j,k)
47              end do
48          end do
49      end do
50      close(10)

```

```

51
52      ! Compute initial total magnetization
53      do i = 1, L
54          do j = 1, L
55              do k = 1, L
56                  M = M + spin(i,j,k)
57              end do
58          end do
59      end do
60
61      ! Compute initial energy using periodic boundary
        conditions
62      do i = 1, L
63          do j = 1, L
64              do k = 1, L
65                  a = i + 1; if (i == L) a = 1
66                  b = i - 1; if (i == 1) b = L
67                  c = j + 1; if (j == L) c = 1
68                  d = j - 1; if (j == 1) d = L
69                  ee = k + 1; if (k == L) ee = 1
70                  f = k - 1; if (k == 1) f = L
71
72                  E = E - J_ising * real(spin(i,j,k) * ( &
73                      spin(a,j,k) + spin(b,j,k) + spin(
74                          i,c,k) + &
75                          spin(i,d,k) + spin(i,j,ee) + spin
76                              (i,j,f) ))
77              end do
78          end do
79      end do
80      E = E * 0.5d0      ! Correct for double counting
81
82      mag = M / real(N)
83      print*, 'Initial_Energy_E_per_spin=', E, E/real(N
84          )
85      print*, 'Initial_Magnetization_M_per_spin=', M,

```

```

mag
83
84      ! Open output file for temperature loop results
85      open(11, file='T_loop_L32.dat', status='replace')
86
87      ! Loop over the temperature range (T_temp in integer;
      !   convert to real T)
88      do T_temp = Tf, Ti, dT
89          T = dfloat(T_temp) / 100.0d0  ! E.g., 400 becomes
          4.00
90
91          ! Reset accumulators for the current temperature
92          av_m    = 0.0d0
93          av_e    = 0.0d0
94          av_m_N  = 0.0d0
95          av_e_N  = 0.0d0
96          av_m2   = 0.0d0
97          av_m4   = 0.0d0
98          av_e2   = 0.0d0
99          av_abs_m = 0.0d0
100         ! Perform niter Monte Carlo sweeps.
101         ! Each "sweep" consists of  $L^3$  single-spin update
          attempts.
102         do time = 1, niter
103             do mm = 1, L
104                 do nn = 1, L
105                     do oo = 1, L
106                         ! Choose a random lattice site
107                         call random_number(r)
108                         i = int(r * real(L)) + 1
109                         call random_number(q)
110                         j = int(q * real(L)) + 1
111                         call random_number(h)
112                         k = int(h * real(L)) + 1
113
114                         ! Determine neighbors with

```

```

115         periodic boundary conditions
116     a = i + 1; if (i == L) a = 1
117     b = i - 1; if (i == 1) b = L
118     c = j + 1; if (j == L) c = 1
119     d = j - 1; if (j == 1) d = L
120     ee = k + 1; if (k == L) ee = 1
121     f = k - 1; if (k == 1) f = L
122
123     ! Calculate energy before the
124         trial flip
125     Ei = -J_ising * real( spin(i,j,k)
126         * ( &
127             spin(a,j,k) + spin(b,j,k)
128             + spin(i,c,k) + &
129             spin(i,d,k) + spin(i,j,ee
130                 ) + spin(i,j,f) ) )
131
132     ! Perform the trial flip
133     spin(i,j,k) = -spin(i,j,k)
134     ! Calculate energy after the flip
135     Ef = -J_ising * real( spin(i,j,k)
136         * ( &
137             spin(a,j,k) + spin(b,j,k)
138             + spin(i,c,k) + &
139             spin(i,d,k) + spin(i,j,ee
140                 ) + spin(i,j,f) ) )
141
142     dE = Ef - Ei
143
144     ! Metropolis acceptance/rejection
145     if (dE <= 0.0d0) then
146         E = E + dE
147         M = M + (2.0d0 * real(spin(i,
148             j,k)))
149     else
150         u = exp(-dE / T)
151         call random_number(h)
152         if (h < u) then

```

```

142         E = E + dE
143         M = M + (2.0d0 * real(
144             spin(i,j,k)))
145     else
146         ! Reject the trial flip:
147         ! revert the spin
148         spin(i,j,k) = -spin(i,j,k)
149     )
150 end if
151 end if
152 end do
153 end do
154
155 ! After equilibration, accumulate data every
156 n_stat sweeps
157 if (time > n_equil) then
158     if (mod(time, n_stat) == 0) then
159         mag = M / real(N) !
160         ! Magnetization per spin
161         abs_mag = abs(M) / real(N) !
162         ! Absolute magnetization per spin
163         av_m = av_m + mag
164         av_abs_m = av_abs_m + abs_mag
165         av_e = av_e + E / real(N)
166         av_m_N = av_m_N + M
167         av_e_N = av_e_N + E
168         av_m2 = av_m2 + (M * M)
169         av_m4 = av_m4 + (M * M * M * M)
170         av_e2 = av_e2 + (E * E)
171     end if
172 end if
173 end do ! End of Monte Carlo sweeps
174
175 ! Calculate the number of measurement points
176 n_meas = (niter - n_equil) / n_stat

```

```

172         if (n_meas <= 0) then
173             print*, "Error: No measurement points
                    collected. Increase niter or decrease
                    n_equil."
174             stop
175         end if
176
177         ! Compute the averages (per measurement point)
178         av_m = av_m / dfloat(n_meas)
179         av_abs_m = av_abs_m / dfloat(n_meas)
180         av_e = av_e / dfloat(n_meas)
181
182         ! Compute the variance-based estimates (specific
                    heat and susceptibility)
183         cv = ((av_e2 / dfloat(n_meas)) - ((av_e_N /
                    dfloat(n_meas))**2)) / (T * T)
184         chi = ((av_m2 / dfloat(n_meas)) - ((av_m_N /
                    dfloat(n_meas))**2)) / T
185         binder = 1.0d0 - (av_m4 / dfloat(n_meas)) / (3.0
                    d0 * ((av_m2 / dfloat(n_meas))**2))
186         ! Write the results for the current temperature:
187         write(11,*) T, av_m, av_e, cv, chi, av_abs_m,
                    binder
188     end do ! End temperature loop
189
190     close(11)
191     print*, "Final values (for last temperature):"
192     print*, "Magnetization per spin=", av_m
193     print*, "Energy per spin=", av_e
194     print*, "Specific heat=", cv
195     print*, "Susceptibility=", chi
196     print*, "Absolute value Magnetization per spin=",
        av_abs_m
197     print*, "Binder's cumulant=", binder
198     deallocate(spin)
199 end program mcs_fixed

```

Listing A.1: Source code for `mcs_fixed`